
Head-Level Spatial Reasoning Analysis and Targeted Attention Reallocation for Vision-Language Models

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Abstract

We present a mechanism-driven study of spatial reasoning in vision-language models and use the findings to design a stronger attention intervention strategy. First, we analyze layer-17 head behavior with YOLO-grounded overlap and overlap-to-correctness AUROC on Controlled_Images_A, and find a strong negative relation between “spatially concentrated” heads and “functionally useful” heads. This suggests that final correctness is often supported by heads with weaker current spatial focus but higher latent functional potential. Second, we test a naive in-box boosting idea and show that directly forcing attention into detector regions is unstable, motivating a smoother design. We then introduce margin-guided continuous attention reallocation, which controls intervention strength by decoding confidence margin and redistributes image attention without hard clipping. Finally, we evaluate on four benchmarks (COCO_QA_one_obj, COCO_QA_two_obj, VG_QA_one_obj, VG_QA_two_obj) and observe consistent gains on harder two-object settings while preserving one-object performance. Code: https://github.com/yinhhhh/spatial_reasoning_head.

1 Introduction

Vision-language models (VLMs) have achieved strong performance across multimodal tasks [Liu et al., 2023, Bai et al., 2023], yet spatial reasoning remains fragile, especially when fine-grained object relations must be inferred from visual evidence. A common line of work addresses this issue by adjusting image-side attention during decoding, with the goal of improving visual grounding and reducing relation errors [Anonymous, 2026]. While such attention reweighting methods are increasingly used, most studies treat attention heads as a homogeneous resource and focus on aggregate intervention strength.

We argue that this head-agnostic view misses an important mechanistic question: *do different heads contribute unequally to spatial reasoning?* If so, intervention should not only decide *how much* visual attention to allocate, but also *where in head space* this allocation should be concentrated. This perspective motivates our central objective: to characterize head-level functional differences in spatial reasoning and leverage them to improve downstream spatial performance.

Our core research question is:

Can we identify and reallocate spatial reasoning responsibility across heads in a way that improves spatial grounding while preserving semantic correctness?

To answer this question, the paper follows a three-stage narrative:

1. **Head analysis:** establish that head-level spatial roles are real and measurable, and identify which heads are already spatially effective.

2. **Improved allocation method:** design a better strategy for distributing image attention, moving beyond naive or purely global reweighting.
3. **Mechanism-backed validation:** use the intervention outcomes to show that heads with latent but underutilized spatial-attention potential are key targets for improving spatial reasoning.

2 Preliminaries

Attention. Transformer attention maps an input token sequence to contextualized representations through query, key, and value projections [Vaswani et al., 2017]. Given token features $X \in \mathbb{R}^{n \times d}$, each head computes

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad (1)$$

and attention weights

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right), \quad (2)$$

followed by head output

$$Z = AV. \quad (3)$$

Multi-head attention concatenates head outputs and applies a final linear projection.

In autoregressive decoding, the model produces next-token logits z_t at step t , and token probabilities are obtained by

$$p(y_t = i \mid y_{<t}, x) = \frac{\exp(z_{t,i})}{\sum_j \exp(z_{t,j})}. \quad (4)$$

The generated token is selected from this distribution (e.g., greedy or sampling decoding).

Vision-Language Models (VLMs). Models such as LLaVA and Qwen-VL [Liu et al., 2023, Bai et al., 2023] process an image by first encoding it with a vision backbone (typically ViT-style), which divides the image into patches and converts them into a sequence of visual features (visual tokens). These visual tokens are then projected into the language model hidden space through a connector/projection module. Finally, the visual tokens are concatenated with text tokens (prompt tokens) and fed into the decoder, so self-attention can jointly model image-text interactions in one token sequence. Our later method sections build on this standard multimodal tokenization-and-fusion pipeline.

3 Head-Level Spatial Analysis

Before intervention design, we first identify which heads are spatially active and which heads are functionally tied to correctness.

Experiment settings. We conduct the head-level analysis on `Controlled_Images_A`, a synthetic and tightly controlled benchmark used in AdaptVis-style evaluation. The images are human-constructed to isolate spatial relations (e.g., left/right/on/under) with limited visual confounders, which makes it suitable for diagnosing whether attention heads truly capture spatial cues. The question set follows the AdaptVis protocol [Anonymous, 2026]: GPT-4-generated relation questions in a fixed-choice format, such as “Where is the object in relation to the anchor object?” with options {left, right, on, under}. In the AdaptVis analysis, layer 17 attention is reported as most correlated with generation behavior; therefore, we focus our head-level analysis on layer 17 and use the single-box filtering protocol for reliable region attribution.

3.1 Head-wise attention distribution (YOLO overlap)

We first examine where each head places attention mass by computing YOLO-grounded overlap, with the following concrete procedure. For each sample, we run YOLO [Redmon et al., 2016] to obtain the object box and retain single-box cases to reduce ambiguity. The box is expanded by a fixed scale ($1.2\times$ in our current setting) and rasterized into a binary mask $M \in \{0, 1\}^{H \times W}$. For each head h , we extract its attention map, project/resize it to the same grid, and normalize it into a

heatmap H_h such that $\sum_{i,j} H_h(i,j) = 1$. We then compute

$$\text{Overlap}_h = \frac{\sum_{i,j} H_h(i,j) M(i,j)}{\sum_{i,j} H_h(i,j)}. \quad (5)$$

Intuitively, this is the fraction of a head’s attention mass that falls inside the detector-supported object region. Figure 1 visualizes a 32-head comparison for a representative sofa/armchair sample, showing strong cross-head heterogeneity in spatial focus.

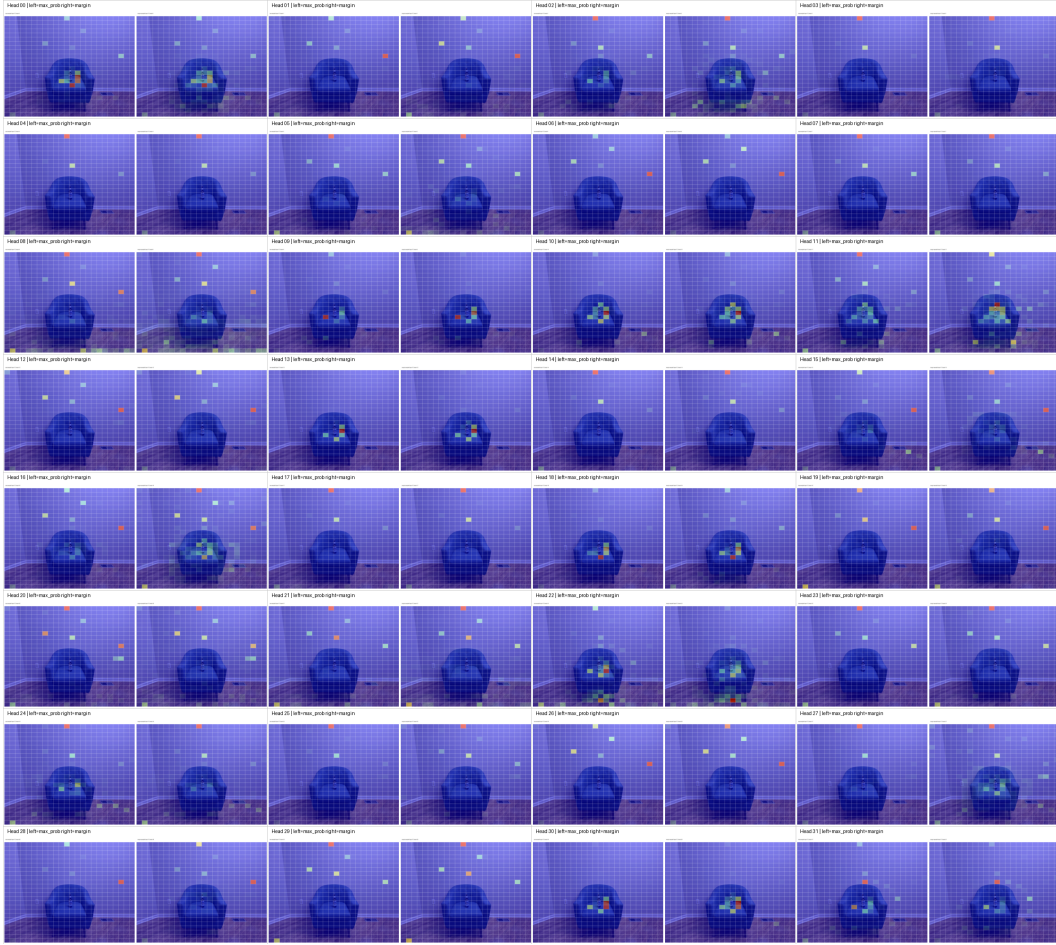


Figure 1: 32-head qualitative comparison on a representative controlled sample (layer 17).

At dataset level, overlap ranking is dominated by heads $\{2, 18, 0, 30, 9, 11, 13, 10\}$, as shown in Figure 2.

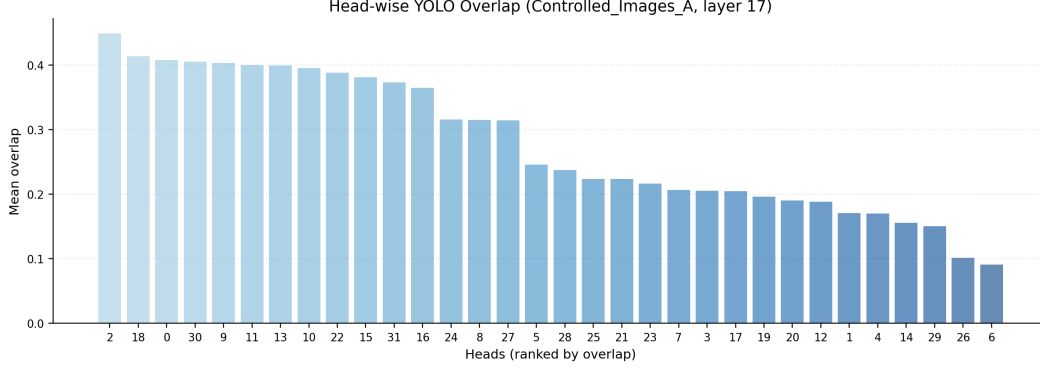


Figure 2: Per-head overlap on Controlled_Images_A under the single-box protocol (heads ranked by overlap; top-8 highlighted).

3.2 AUROC-based functional ranking

Overlap alone does not tell us whether a head helps the final decision. We therefore evaluate each head by AUROC, using per-sample overlap scores as the predictor and answer correctness (0/1) as the label. Concretely, for each head h , we collect $\{\text{Overlap}_h(x), y(x)\}_{x \in \mathcal{D}}$, where $y(x) = 1$ indicates a correct answer, and compute

$$\text{AUROC}_h = \text{AUC}(\text{Overlap}_h(x) \rightarrow y(x)). \quad (6)$$

Top AUROC heads are $\{14, 29, 26, 6, 7, 4, 20, 17\}$ (contrast visible in Figures 3 and 4), which is almost disjoint from the top-overlap group.

This contrast is not anecdotal: overlap and AUROC are strongly negatively correlated across heads (Pearson $r \approx -0.915$), shown in Figures 3 and 4. Figure 4 sorts heads from low to high overlap and plots overlap/AUROC as two curves under the same head order, making the inverse tendency visually explicit. This is a surprising result and suggests that heads with already strong spatial concentration are not necessarily the ones most tied to final correctness.

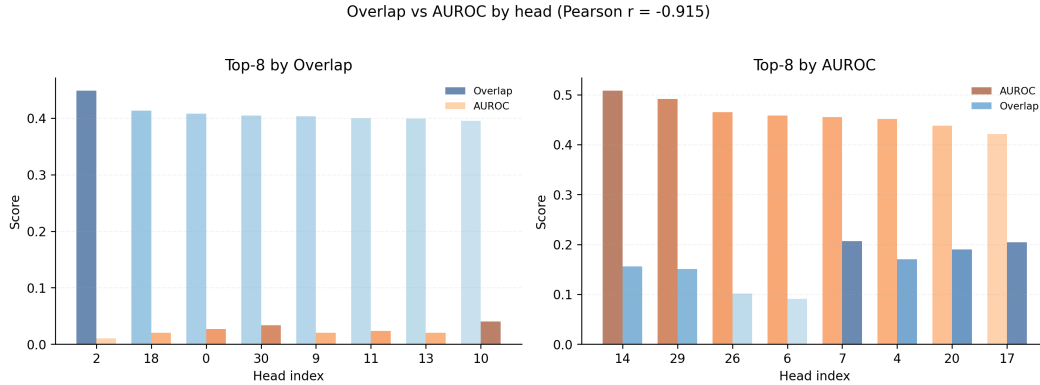


Figure 3: Inverse relation between overlap and AUROC across heads (top-8 overlap vs top-8 AUROC groups).

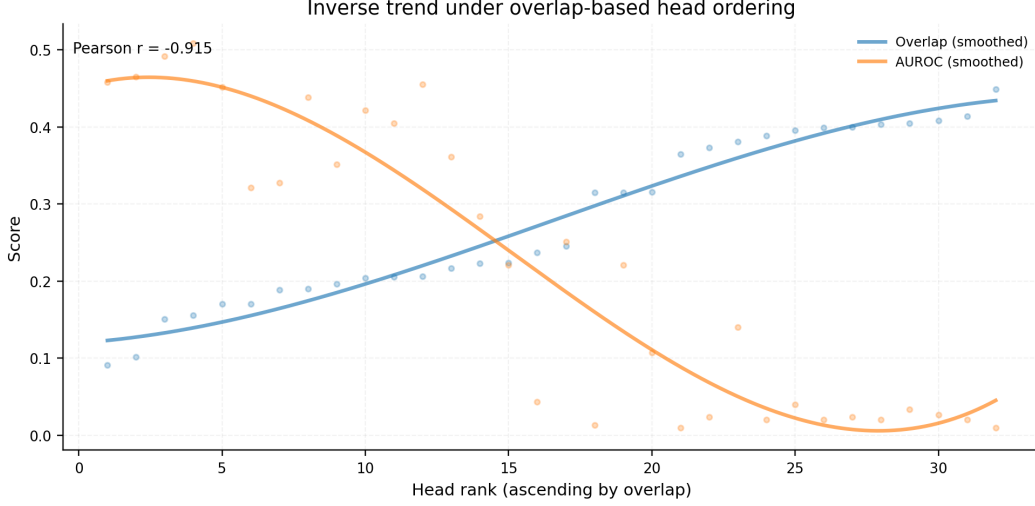


Figure 4: Two-line view of inverse relation: heads are sorted by overlap (small to large), while overlap (blue) and AUROC (orange) are plotted on the same head order.

A correctly solved sample is often supported not by heads with already strong spatial attention, but by heads with weaker current spatial focus yet higher functional potential.

4 Our Method: Function-Aware Spatial Attention Reallocation

Experiment settings. Unless otherwise specified, this section follows the same evaluation setup as Section 3: 11ava1.5, the AdaptVis protocol, and matched 100-sample splits for cross-method comparison. We evaluate on four benchmarks with increasing relational difficulty: COCO_QA_one_obj and VG_QA_one_obj (single-object relation queries), and COCO_QA_two_obj and VG_QA_two_obj (two-object relation queries). This split allows us to check whether a method is robust only on simpler one-object questions or still effective when relational interference increases.

Naive idea. Our first idea was straightforward: directly boost attention inside the YOLO box. If a region detector marks the target object, increasing weight in that region seems likely to help spatial reasoning. In practice, this direct strategy is unstable and often hurts semantic consistency; therefore, we do not include its detailed quantitative results in the main text.

4.1 Margin-Guided Continuous Reallocation

To address the failure mode above, we replace hard region forcing with a confidence-aware and smooth reallocation mechanism.

(i) Margin-guided continuous control. At decoding step t , let $p_{(1),t}$ and $p_{(2),t}$ be top-1 and top-2 token probabilities. We define confidence margin

$$m_t = p_{(1),t} - p_{(2),t}. \quad (7)$$

Intervention strength is then set continuously by

$$\alpha_t = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot \sigma(\beta(m_t - \tau)), \quad (8)$$

where β controls slope and τ controls the transition point.

(ii) Attention reweighting without hard clipping. Given image-side attention A_t , we apply

$$\tilde{A}_t = \text{Normalize}(A_t^{\alpha_t}), \quad (9)$$

which is the core execution step of our method: the confidence-controlled exponent α_t is converted into a smooth redistribution of attention mass over image patches. In practice, $\alpha_t > 1$ sharpens attention to concentrate evidence, while smaller values keep attention diffuse, avoiding abrupt region-level clipping.

4.2 Experimental Results

Table 1 reports the 100-sample comparison on four benchmarks. “LLaVA-1.5” denotes the no-intervention model output, “AdaptVis” denotes the original max-prob/hard setting, and “Ours” denotes the best margin+continuous setting under the same evaluation protocol.

Table 1: Accuracy comparison on four datasets (100-sample view).

Method	COCO_QA_one_obj	COCO_QA_two_obj	VG_QA_one_obj	VG_QA_two_obj
LLaVA-1.5 (no intervention)	—	—	—	—
AdaptVis	0.54	0.52	0.38	0.45
Ours (margin + continuous)	0.54	0.54	0.38	0.49

Our method preserves performance on one-object settings and improves two-object settings (COCO_QA_two_obj: +0.02; VG_QA_two_obj: +0.04), which is consistent with the hypothesis that better head-level spatial responsibility allocation is most beneficial when relational complexity increases. We leave the “LLaVA-1.5” row as a placeholder to be filled by a no-intervention rerun under the exact same 100-sample split.

5 Conclusion

We provide an analysis-driven workflow: identify spatially useful heads, test targeted interventions, and evaluate trade-offs between spatial alignment and task correctness. The study contributes (i) a concrete head-analysis protocol (overlap + AUROC), (ii) a targeted attention reallocation strategy, and (iii) a set of negative and positive findings that clarify where spatial head reallocation helps or fails.

A Failure Case Discussion

An additional variant we explored is *top-patch flattening*: for the top- ρ fraction of patches with highest attention, enforce local smoothing toward a shared high-attention level:

$$\hat{A}_t(i) = \begin{cases} (1 - \lambda)\tilde{A}_t(i) + \lambda\bar{A}_t^{\text{top}}, & i \in \mathcal{P}_\rho, \\ \tilde{A}_t(i), & \text{otherwise.} \end{cases} \quad (10)$$

Although this variant was intended to stabilize redistribution, it often over-flattened critical contrast and degraded answer quality (especially on two-object settings) in our current runs. We therefore treat top-patch flattening as a failure-case direction rather than part of the main method.

B Best Hyperparameter Selection (100-sample)

Table 2 summarizes the selected best hyperparameters from the 100-sample tuning summary (baseline_vs_best_4ds_100.md). We report the final configuration used for each dataset under the margin+continuous framework.

Table 2: Best hyperparameters selected from 100-sample tuning.

Dataset	Option	w_1/w_2	Threshold	Criterion	Adapt weighting	Token/Image settings
COCO_QA_one_obj	four	0.5 / 1.2	0.08	margin	sigmoid ($\alpha=6$, span=0.1)	window=3, agg=mean, image_scale=1.0
COCO_QA_two_obj	four	0.5 / 1.2	0.12	margin	sigmoid ($\alpha=6$, span=0.1)	window=3, agg=mean, image_scale=1.3
VG_QA_one_obj	six	0.5 / 2.0	0.32	margin	sigmoid ($\alpha=6$, span=0.1)	window=3, agg=mean, image_scale=1.0
VG_QA_two_obj	six	0.5 / 2.0	0.24	margin	sigmoid ($\alpha=6$, span=0.1)	window=3, agg=mean, image_scale=1.0

C Appendix: Naive-Idea Evaluation (TODO)

We plan to add a dedicated quantitative comparison for the naive YOLO in-box boosting idea in a future revision. The appendix table will report four-dataset results under the same 100-sample split as the main method, including accuracy deltas versus AdaptVis and representative failure examples.

References

- Anonymous. Adaptvis: Adaptive attention reweighting for visual reasoning in multimodal language models. *arXiv preprint arXiv:XXXX.XXXXX*, 2026. Please replace with the exact bibliographic entry.
- Jinze Bai, Shuai Bai, Shusheng Yang, Shugang Wang, Siling Tan, Ke Wang, Xinhui Li, Rulin Hu, Dian Ren, Xia Zhang, Chuan Zhou, Wen Ge, Junyang Lin, Yichang Sun, Junsong Chen, Jiaxuan Li, Guosheng Guo, Nan Ding, Yujia Zhang, Xuan Li, Dayiheng Li, Tianyu Liu, Jun Zhu, Yi Ma, and Xipeng Qiu. Qwen-vl: A frontier large vision-language model with versatile abilities. *arXiv preprint arXiv:2308.12966*, 2023.
- Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *arXiv preprint arXiv:2304.08485*, 2023.
- Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 779–788, 2016.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.